

MIPS Compiler Backend: Design and Implementation

CS 4240: Compilers and Interpreters, Project 2

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1 High-Level Architecture and Algorithm Choices

1.1 Pipeline Organization

Our backend is organized as a pipeline with clear phase boundaries: IR ingestion and CFG construction, instruction selection, register allocation, and assembly emission. This separation is intentional and provides several critical benefits for the development process. Each phase has a single responsibility, exposes an explicit contract to the next phase, and can be validated in isolation. In practice, this modularity kept the engineering risk low and made it easy to iterate: when an error/problem appeared, we could often bisect to a single phase by substituting the naive allocator, printing the selected virtual MIPS, or running a block-local liveness sanity check.

The transformation pipeline follows a strict sequence that ensures correctness at each stage:

IR Instructions → Basic Blocks → Virtual MIPS → Physical MIPS → Final Assembly

This linear flow allows us to insert validation points between phases, catching errors early before they propagate through the system. The modular design also enables targeted optimization of individual phases without disrupting the overall architecture.

1.2 Virtual Register Design

A central design decision is that instruction selection produces a virtualized MIPS program that never mentions physical registers. Every variable and temporary is named with a structured virtual naming scheme.

Listing 1: Virtual Register Naming Convention

1	<code>\$v-local--x</code>	# Local variable x
2	<code>\$v-param--y</code>	# Parameter y
3	<code>\$v-temp--off</code>	# Temporary for offset calculation
4	<code>\$v-temp--addr</code>	# Temporary for address computation
5	<code>\$v-temp--base</code>	# Temporary for base address
6	<code>\$v-temp--tmp</code>	# General temporary

The structured naming makes the IR→MIPS mapping deterministic and easy to diff, which was invaluable for debugging array addressing and branch lowering. When examining generated code, we can immediately identify the purpose of each virtual register, streamlining the debugging process.

1.3 Register Allocation

We implemented two register allocation algorithm. This design allows us to validate optimizations against a known-correct baseline(naive) while providing performance(intrablock) improvements when desired.

1.3.1 Naive Allocator

The naive allocator: at every instruction boundary, all virtuals live in memory. It generates verbose code with many `lw/sw` instructions but is easy to reason about and verify for

correctness. This baseline served for both correctness and performance testing throughout development.

Listing 2: Naive Allocator Core Logic

```

1 for ins in virtMips:
2     # Load phase: bring all operands into registers
3     for operand in ins.operands:
4         emit("lw $t{}, {}($fp)", reg_num, offset[operand])
5
6     # Execute phase: perform operation
7     emit(ins.with_physical_registers())
8
9     # Store phase: save result to memory
10    if ins.has_destination:
11        emit("sw $t{}, {}($fp)", dest_reg, offset[dest])

```

1.3.2 Intra-Block Greedy Allocator

The intra-block greedy allocator uses local liveness and a "farthest next-use" eviction policy to keep hot values in a small cache (\$t1–\$t4) inside each basic block. It inserts loads on first use and stores on last definition, dramatically reducing memory traffic in loop-dominated code. The allocator flushes all dirty values at block boundaries and before jal/syscall instructions, ensuring blocks remain independent and calling conventions are respected.

Listing 3: Greedy Allocator Spill Decision

```

1 function chooseVictim(i, nextUse, regMap, dirty):
2     best_victim = null
3     best_next_use = -1
4
5     for (virt, phys) in regMap:
6         nu = nextUse[i].get(virt, infinity)
7         is_dirty = virt in dirty
8
9         # Perfect victim: dead and clean
10        if nu == infinity and not is_dirty:
11            return virt # Immediate eviction
12
13        # Better victim: used later or clean
14        if nu > best_next_use:
15            best_victim = virt
16            best_next_use = nu
17
18    return best_victim

```

This policy captures a large fraction of the available benefit on loop-dominated code without the complexity of global allocation. The greedy approach typically reduces memory operations by 40-60% compared to the naive register allocation.

1.4 Assembly and Frame Management

We use a conventional MIPS frame with aligned allocation, save/restore of \$fp/\$ra, and a clean epilogue before `jr $ra`. The emission phase ensures strict compliance with MIPS conventions.

Listing 4: Function Prologue/Epilogue

```
1 function_name:
2     # Prologue
3     addi $sp, $sp, -frameSize      # Allocate frame
4     sw    $ra, frameSize-4($sp)    # Save return address
5     sw    $fp, frameSize-8($sp)    # Save frame pointer
6     addi $fp, $sp, frameSize-8     # Establish new frame
7
8     # Function body here
9
10    # Epilogue
11    lw     $ra, frameSize-4($sp)    # Restore return address
12    lw     $fp, frameSize-8($sp)    # Restore frame pointer
13    addi $sp, $sp, frameSize        # Deallocate frame
14    jr     $ra                     # Return
```

2 Low-Level Design Decisions and Rationale

2.1 Integer Arithmetic Implementation

Integer arithmetic maps 1:1 to MIPS operations wherever possible to minimize instruction count and maintain semantic clarity. We use `addi` when an immediate fits in 16 bits, falling back to `li` for full-width constants that require multiple instructions.

Listing 5: Arithmetic Instruction Selection

```
1 case ADD(dst, a, b):
2     if is_immediate(b) and fits_in_16_bits(b):
3         # Optimize to single instruction
4         emit("addi v({}), v({}), {}", dst, a, b)
5     elif is_immediate(b):
6         # Need temporary for large constant
7         emit("li v(temp--imm), {}", b)
8         emit("add v({}), v({}), v(temp--imm)", dst, a)
9     else:
10        # Register-to-register operation
11        emit("add v({}), v({}), v({})", dst, a, b)
```

2.2 Array Addressing

For arrays, we standardized on a single addressing template that ensures consistency across all array operations.

Listing 6: Array Address Calculation Template

```

1 # Step 1: Compute byte offset (idx * 4)
2 sll v(temp--off), v(idx), 2
3
4 # Step 2: Form base address
5 # For local arrays:
6 addi v(temp--base), $fp, frameOffset(arr)
7 # For parameter arrays (pointers):
8 move v(temp--base), v(arr_ptr)
9
10 # Step 3: Calculate element address
11 sub v(temp--addr), v(temp--base), v(temp--off)
12
13 # Use the address
14 lw v(dst), 0(v(temp--addr)) # For loads
15 sw v(src), 0(v(temp--addr)) # For stores

```

By reserving named temporaries (`$v-temp--off`, `$v-temp--base`, `$v-temp--addr`) for each step, the resulting instruction patterns are visually obvious and uniform across the codebase. This approach eliminated an entire class of array indexing bugs during development.

2.3 Control Flow Implementation

Control flow is mapped to explicit branch instructions with function-scoped labels to prevent naming conflicts across functions.

Listing 7: Branch Instruction Lowering

```

1 case BRLT(label, a, b):
2     # Generate function-scoped label
3     scoped_label = "{}_{}".format(label, current_function)
4
5     if is_immediate(b):
6         # Load immediate to temporary
7         emit("li v(temp--cmp), {}", b)
8         emit("blt v({}), v(temp--cmp), {}", a, scoped_label)
9     else:
10        # Direct register comparison
11        emit("blt v({}), v({}), {}", a, b, scoped_label)

```

This straightforward approach ensures that control flow is easy to trace in the generated assembly.

2.4 Calling Convention Implementation

For Tiger, we use a small, practical calling convention: up to four integer arguments go in `a0-a3`; the return value comes back in `v0`; and `fp` provides a stable frame for locals. The allocator is conservative around `jal` and `syscall`: it spills dirty registers before the call and reloads them after.

Listing 8: Function Call

```

1 function lowerCall(target, args, fn):
2     if fn.parameters.size > 0:
3         emit("addi $sp, $sp, -{}", 4 * fn.parameters.size)
4         for i in 0..fn.parameters.size-1:
5             emit("sw $a{}", {}($sp)", i, i*4)
6
7     for i, arg in enumerate(args):
8         if i < 4:
9             if is_immediate(arg):
10                emit("li $a{}", {}, i, arg)
11            else:
12                emit("move $a{}", v({}), i, arg)
13        else:
14            emit("move v(stackarg--{}), v({})", i-4, arg)
15
16    # Save context and call
17    emit("addi $sp, $sp, -8")
18    emit("sw $fp, 0($sp)")
19    emit("sw $ra, 4($sp)")
20    emit("jal {}", target)
21
22    # Restore context
23    emit("lw $fp, 0($sp)")
24    emit("lw $ra, 4($sp)")
25    emit("addi $sp, $sp, 8")
26
27    # Restore caller's registers
28    if fn.parameters.size > 0:
29        for i in 0..fn.parameters.size-1:
30            emit("lw $a{}", {}($sp)", i, i*4)
31            emit("addi $sp, $sp, {}", 4 * fn.parameters.size)

```

2.5 Stack Layout Design

The stack frame starts at offset -8 to reserve space for the saved `$fp` (at -8) and `$ra` (at -4), mirroring the prologue/epilogue that saves/loads `$ra` and `$fp` and sets `$fp = $sp + (stackSize - 8)`. Each parameter and local receives a fixed “home” offset from `$fp`: scalars use 4 bytes, arrays use $4 \times \text{length}$ bytes laid out contiguously, and offsets grow negatively ($-8, -12, -16, \dots$) so every variable has one constant address `Addr(offset, $fp)`. After assigning homes, we compute the total frame size and round it up to an 8-byte boundary for alignment. This design matters because fixed homes let both allocators reliably spill/reload with `lw/sw` using the known `Addr(offset, $fp)`; contiguous arrays allow a standard address pattern (shift index by 2, form the base from `$fp + array_offset`, then compute the element address); and a known, aligned `totalStackSize` guarantees a correct prologue/epilogue to save/restore `$fp/$ra` and adjust `$sp`.

`Addr(offset, $fp)` for a spill or reload.

Listing 9: Stack Layout Calculation

```

1 function buildStackLayout(function):
2     offset = -8 # Reserve for $fp and $ra
3     offsets = {}
4
5     # Allocate parameters
6     for param in function.parameters:
7         offsets[param.name] = offset
8         offset -= 4
9
10    # Allocate local scalars
11    for var in function.locals:
12        if not var.is_array:
13            offsets[var.name] = offset
14            offset -= 4
15
16    # Allocate arrays
17    for var in function.locals:
18        if var.is_array:
19            offsets[var.name] = offset
20            offset -= 4 * var.size
21
22    # Align frame to 8 bytes
23    frame_size = -offset + 8
24    if frame_size % 8 != 0:
25        frame_size += 4
26
27    return (offsets, frame_size)

```

3 Software Engineering Challenges and Resolutions

3.1 Phase Boundary Enforcement

Instruction selection accidentally emits physical registers, which then restricts the register allocator. We prevented this with a strict rule: selection may emit only virtual registers; physical registers are introduced only during allocation. This rule surfaced early bugs where address math hard-coded \$t8/\$t9 instead of using virtual temporaries (\$v-temp--).

3.2 Array Addressing Errors

Array bugs are hard to spot because they usually corrupt data instead of crashing. Early on, we had problems when computing negative offsets from the frame pointer.

4 Testing and Validation Strategy

4.1 Correctness Testing Framework

We test each benchmark in both modes (`--naive` and `--greedy`) and use the naive allocator as a benchmark. For performance, we run under the SPIM interpreter and

count dynamic `lw/sw` instructions, branches, and instruction totals.

4.2 Benchmark Performance Analysis

The following tables show the performance for quicksort and prime, the two given test cases. It shows the Naive and Greedy register allocation algorithms. Additionally, it shows the reduction in loads as well.

Table 1: PRIME benchmark performance comparison between Naive and Block implementations

File	Instructions	Reads	Writes	Branches	Other
NAIVE					
0.in	44	6	10	3	25
1.in	179	60	36	15	68
2.in	312	112	63	27	110
3.in	445	164	90	39	152
4.in	286	100	57	26	103
5.in	625	232	126	57	210
6.in	2174	840	441	195	698
7.in	286	100	57	26	103
8.in	8159	3180	1656	735	2588
9.in	16139	6300	3276	1455	5108
BLOCK					
0.in	41	3	10	3	25
1.in	145	29	33	15	68
2.in	247	53	58	27	109
3.in	349	77	83	39	150
4.in	228	47	53	26	102
5.in	489	109	117	57	206
6.in	1675	389	408	195	683
7.in	228	47	53	26	102
8.in	6265	1469	1533	735	2528
9.in	12385	2909	3033	1455	4988

QUICKSORT Benchmark - Detailed Metrics

Table 2: QUICKSORT benchmark performance comparison between Naive and Block implementations

File	Instructions	Reads	Writes	Branches	Other
NAIVE					
0.in	2686	1088	501	179	918
1.in	6125	2551	1127	405	2042
2.in	9776	4117	1777	647	3235
3.in	13574	5760	2458	890	4466
4.in	16895	7164	3057	1107	5567
5.in	22203	9555	3979	1449	7220
6.in	25038	10717	4481	1631	8209
7.in	29622	12744	5276	1936	9666
8.in	33351	14351	5966	2186	10848
9.in	37398	16116	6660	2440	12182
BLOCK					
0.in	2235	646	492	179	918
1.in	5072	1517	1108	405	2042
2.in	8080	2450	1748	647	3235
3.in	11184	3409	2419	890	4466
4.in	13917	4235	3008	1107	5567
5.in	18236	5647	3920	1449	7220
6.in	20556	6304	4412	1631	8209
7.in	24318	7519	5197	1936	9666
8.in	27415	8504	5877	2186	10848
9.in	30686	9503	6561	2440	12182

Statistical Summary

Table 3: Instruction count reduction from Naive to Block optimization

Benchmark	Naive Instructions	Block Instructions	Reduction
PRIME (Total)	28,649	21,852	23.7%
QUICKSORT (Total)	196,668	161,603	17.8%

Table 4: Read operation reduction from Naive to Block optimization

Benchmark	Naive Reads	Block Reads	Reduction
PRIME (Total)	11,094	5,132	53.7%
QUICKSORT (Total)	82,103	49,734	39.4%

Table 5: Write operation reduction from Naive to Block optimization

Benchmark	Naive Writes	Block Writes	Reduction
PRIME (Total)	5,812	5,381	7.4%
QUICKSORT (Total)	34,282	34,742	-1.3%

Table 6: Average performance metrics across all test cases

Benchmark	Metric	Naive Avg	Block Avg	Improvement
PRIME	Instructions	2,865	2,185	23.7%
	Reads	1,109	513	53.7%
	Writes	581	538	7.4%
	Branches	158	158	0.0%
QUICKSORT	Instructions	19,667	16,160	17.8%
	Reads	8,210	4,973	39.4%
	Writes	3,428	3,474	-1.3%
	Branches	1,487	1,487	0.0%

Key Findings

- **PRIME benchmark:** Block mode cuts total instructions by 23.7% and reads by 53.7% vs. Naive.
- **QUICKSORT benchmark:** Block mode cuts total instructions by 17.8% and reads by 39.4%.
- **Read efficiency:** Both benchmarks show large read reductions (PRIME is > 50%).
- **Writes:** PRIME writes drop 7.4%; QUICKSORT writes rise 1.3% gains mainly come from fewer reads.
- **Branches:** Branch counts are unchanged, so control flow isn't affected.

4.3 Build and Execution Scripts

Scripts (`build.sh`, `run.sh`) give repeatable builds and a consistent CLI that matches the spec. `run.sh` follows grading rules: with two args (`file.ir`, `mode`) it writes `out.s` next

to the input; with three args it writes to the given output path. Internally, it calls the main compiler class to run the pipeline with the chosen allocator mode.

Listing 10: Run Script Implementation

```

1 #!/bin/bash
2 # run.sh - Compiler invocation script
3
4 if [ $# -eq 2 ]; then
5     # Two arguments: input and mode
6     INPUT="$1"
7     MODE="$2"
8     OUTPUT="$(dirname "$INPUT")/out.s"
9 elif [ $# -eq 3 ]; then
10    # Three arguments: input, mode, output
11    INPUT="$1"
12    MODE="$2"
13    OUTPUT="$3"
14 else
15     echo "Usage: $0 <input.ir> <--naive|--greedy> [output.s]"
16     exit 1
17 fi
18
19 # Validate mode
20 if [ "$MODE" != "--naive" ] && [ "$MODE" != "--greedy" ]; then
21     echo "Error: Mode must be --naive or --greedy"
22     exit 1
23 fi
24
25 # Run compiler
26 java -cp build/ Compiler "$INPUT" "$MODE" "$OUTPUT"

```

Our implementation avoids toolchain-specific extensions and depends only on standard Java features.

5 Correctness and Performance Analysis

5.1 Correctness Invariants

Correctness comes from three fundamental invariants that are maintained throughout the compilation process. First, selection uses only virtual registers and preserves IR semantics by direct lowering to MIPS operations with well-formed address expressions. Second, allocation preserves the memory truth: the naive allocator enforces "all virtuals live in memory" at every boundary while the intra-block allocator enforces "at block boundaries and control transfers, all dirty values are spilled." Third, emission creates a valid MIPS frame with proper save/restore sequences, ensuring returns restore `$fp/$ra/$sp` before `jr $ra`.

These invariants are verified through assertion checks at phase boundaries during development and through differential testing against the naive allocator in production.

The systematic enforcement of these invariants eliminated entire categories of bugs that plagued earlier prototypes.

5.2 Performance Characteristics

The performance improvements from the greedy allocator stem from its ability to exploit temporal locality within basic blocks. The allocator identifies and caches the working set of variables that are repeatedly accessed within a block. For typical loop bodies, this working set is small (3-5 variables) and fits entirely in our allocated temporary registers.

The farthest next-use eviction policy ensures that the most valuable registers are retained, approximating optimal offline caching for the known instruction sequence within each block. While this local optimization doesn't capture cross-block reuse opportunities, it delivers most of the available performance benefit with manageable implementation complexity.

5.3 Scalability Considerations

The system handles large programs well—even those with hundreds of basic blocks and thousands of instructions. The intra-block allocator runs in $O(n \times r)$, where n is the number of instructions and r is the number of registers. Memory is mainly used by the next-use tables at $O(n \times v)$, where v is the number of variables, which remains reasonable for complex code.

Because the architecture is modular, we can add new optimization passes or swap in different allocators without disrupting the pipeline. Future optimization, like global allocation or peephole optimizations, can be layered on top of the same codebase.

6 Known Limitations and Future Work

6.1 Current Limitations

Our optimization scope is intra-block only, which means we miss opportunities for register reuse across block boundaries. While this already shrinks dynamic load/store counts significantly on local loops, it doesn't exploit cross-block reuse at join points or loop headers.

Additionally, we perform no peephole optimization. The generated code contains redundant moves, repeated address calculations, and other inefficiencies that could be eliminated with a simple post-processing pass.

7 Conclusion

Our backend is a solid, extensible base for generating MIPS from Tiger IR. It stays correct under the SPIM interpreter by enforcing simple rules at block and function boundaries. With two allocator modes, we get a dependable correctness baseline and faster code on array-heavy loops.

Consistent instruction-selection patterns and cautious handling of tricky cases made the implementation robust across the full Tiger IR. Looking ahead, we can add regional/global register allocation and peephole optimizations without breaking the current design, because the same contracts still apply.